BT101 – Engineering Chemistry

Unit – I

- **Main**
  - Water
  - Hardness
  - Alkalinity
- **Detail**
  - Sources – impurities – hardness – units
  - EDTA method
  - Alkalinity determination – numerical problems

Unit – II

- **Main**
  - Boiler Problems
  - Softening Methods
- **Detail**
  - Sludge – scale – priming – foaming
  - Boiler corrosion – caustic embrittlement
  - Lime soda – zeolite – ion exchange

Unit – III

- **Main**
  - Lubricants
  - Lubrication
- **Detail**
  - Mechanism – classification
  - Viscosity – viscosity index
  - Flash – fire – cloud – pour point
  - Acid number – saponification number

Unit – IV

- **Main**
  - Polymers
  - Polymerization
- **Detail**
  - Types – mechanism (free radical – ionic)
  - Thermoplastic – thermosetting
  - PVC – PMMA – Teflon – Nylon – Buna – vulcanization

Unit – V

- **Main**
  - Phase Equilibrium
  - Corrosion
  - Spectroscopy
- **Detail**
  - Water system – Cu–Ag system
  - Types – mechanisms – prevention
  - Electronic – vibrational – rotational spectroscopy

BT102 – Mathematics-I

Unit – I

- **Main**
  - Calculus
  - Mean Value Theorems
- **Detail**
  - Rolle's theorem
  - Taylor – Maclaurin series
  - Partial differentiation – maxima – minima

Unit – II

- **Main**
  - Definite Integrals
  - Beta – Gamma Functions
- **Detail**
  - Limit of sum
  - Surface area – volume
  - Multiple integrals

Unit – III

- **Main**
  - Sequences
  - Series
- **Detail**
  - Convergence tests
  - Power series
  - Fourier series – Parseval theorem

Unit – IV

- **Main**
  - Vector Space

    - Linear Transformations
- **Detail**
    - Subspace – basis
    - Linear dependence – independence

Unit – V

- **Main**
    - Matrices
    - Eigen Values
- **Detail**
    - Rank – consistency
    - Diagonalization
    - Cayley Hamilton theorem

BT103 – English for Communication

Unit – I

- **Main**
  - Grammar
  - Sentence Structure
- **Detail**
  - Articles – prepositions
  - Subject verb agreement
  - Active – passive – reported speech

Unit – II

- **Main**
  - Vocabulary
  - Comprehension
- **Detail**
  - Prefix – suffix
  - Synonyms – antonyms
  - Reading comprehension

Unit – III

- **Main**
  - Communication
  - Barriers
- **Detail**
  - Process – types
  - 7 C's
  - Non-verbal communication

Unit – IV

- **Main**
  - Writing Skills

    - Reports
- **Detail**
    - Planning – drafting – editing
    - Précis – technical description
    - Laboratory – progress report

Unit – V

- **Main**
    - Business Correspondence
    - Resume
- **Detail**
    - Letters – quotations – complaints
    - Resume – email – tender

BT104 – Basic Electrical & Electronics Engineering

Unit – I

- **Main**
  - DC Circuits
- **Detail**
  - Ohm's law – Kirchhoff laws
  - Superposition – Thevenin theorem
  - Mesh – nodal analysis

Unit – II

- **Main**
  - AC Circuits
  - Three Phase
- **Detail**
  - RMS – power factor – impedance
  - Star – delta connections
  - Power measurement

Unit – III

- **Main**
  - Magnetic Circuits
  - Transformer
- **Detail**
  - Inductance – EMF
  - Losses – efficiency
  - OC – SC tests

Unit – IV

- **Main**
  - Electrical Machines
- **Detail**

    - DC – induction – synchronous machine
    - Torque slip characteristics

Unit – V

- Main
    - Basic Electronics
- Detail
    - Logic gates – flip flops
    - Diodes – BJT
    - CE – CB – CC configuration

BT105 – Engineering Graphics

Unit – I

- **Main**
  - Engineering Drawing
- **Detail**
  - Drawing instruments
  - Scales – conic sections

Unit – II

- **Main**
  - Orthographic Projection
- **Detail**
  - Points – lines – planes

Unit – III

- **Main**
  - Projections of Solids
- **Detail**
  - Prism – pyramid – cone

Unit – IV

- **Main**
  - Sections
  - Development
- **Detail**
  - Sectional views
  - Surface development

Unit – V

- **Main**
  - Isometric Projection
  - CAD
- **Detail**
  - Isometric scale
  - Solid – surface modelling

BT106 – Manufacturing Practices

Unit – I

- **Main**
  - Manufacturing Methods
- **Detail**
  - Casting – forming – machining
  - CNC – additive manufacturing

Unit – II

- **Main**
  - Workshop Practice
- **Detail**
  - Machine shop – fitting
  - Carpentry – welding

BT107 – Internship-I

- **Main**
  - Internship
- **Detail**
  - 60 hours
  - Institute level

BT108 – Swachh Bharat / Unnat Bharat Abhiyan

Unit – I

- **Main**
  - Rural Outreach
- **Detail**
  - Swachh Bharat
  - Unnat Bharat Abhiyan